Physics-Informed Neural Networks for Parameter Estimation and Hidden-State Recovery in an SEIRD Epidemic Model with Time-Varying Transmission:

A Rigorous Investigation of the 2014 West African Ebola Outbreak (Sierra Leone)

Group Project: Mathematical Modelling

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Module 1: Biological Motivation, Formulation, Published Models Synthesis & Phase-Space Analysis
Module 2: Mathematical Analysis (R_0, Next-Generation Matrix, DFE Stability, Final Size, Sensitivity)
Module 3: Vectorized Numerical Solvers & Order of Convergence Verification
Module 4: PINN Architecture, Physics Loss & Synthetic Validation (E2–E4)
Module 5: Real-World Application, Baselines, Rolling Forecasts, Ablation & Limitations (E5–E7)

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Abstract

Compartmental epidemic models are foundational for public health planning, yet standard formulations rely on static transmission coefficients (β) that fail to accommodate behavioral adjustments, non-pharmaceutical interventions, and safe burial practices. Furthermore, surveillance networks capture only manifest clinical outcomes (diagnosed cases and hospital mortalities), leaving the incubating latent class and the depleting susceptible host pool unobserved. In this project, we develop and evaluate a Physics-Informed Neural Network (PINN) framework to reconstruct latent states and discover time-varying transmission functions $\beta(t)$ in an SEIRD dynamical system, using the 2014–2016 West African Ebola Virus Disease (EVD) epidemic in Sierra Leone (N = 7,092,113) as a focal case study. In accordance with course guidelines requiring original analysis upon published models, we first present a rigorous comparative synthesis of three benchmark Ebola models (Legrand et al., Rivers et al., and Althaus), deriving exact phase-plane trajectory relations $i(s)$, the theoretical peak infection threshold $i_{\max}$, and the critical intervention bounds $p_c = 1 - 1/R_0$. We derive the basic reproduction number $R_0 = \beta/(\gamma + \mu) \approx 2.00$ via the next-generation matrix, establish disease-free equilibrium stability ($R_0 s^* < 1$), and verify theoretical final-size equations against high-precision ODE simulations. Vectorized forward Euler and classical Runge-Kutta 4 solvers are implemented from scratch and benchmarked against an adaptive DOP853 integrator ($rtol = 3 \times 10^{-14}, atol = 10^{-16}$), demonstrating empirical convergence orders of 1.001 and 4.085 (theoretical: 1 and 4) and conservation of the total population to 1.2×10^{-15}. Our PINN solves the forward problem to a relative error below 2×10^{-3}, and from noisy cumulative cases and deaths alone it recovers a step change in $\beta(t)$ to 11–12%, the death rate μ to 5%, the hidden susceptible fraction to 0.5% and the exposed and

infectious compartments to 6.5%; an ablation shows which design choices make this work. On WHO surveillance data for Sierra Leone, with a strict 75%/25% chronological split of 29 August 2014 – 25 April 2015, the PINN's held-out case relative error is 1.350%, against 28.558% for constant-β least squares and 0.818% for piecewise-β least squares; its death relative error is 11.772%, against 10.694% and 12.884%. Over 8 rolling 4-week forecasts on the whole 2014–15 epidemic (WHO data), its mean absolute case error is 96.8 cases, against 120.9 (constant-β) and 106.5 (piecewise-β); its death forecasts there are less accurate than both baselines' (69.7 against 50.7 and 42.0).

1 Introduction and Biological Motivation

Infectious disease outbreaks are governed by non-linear contact dynamics, pathogen virulence, and host behavior [1, 2]. The 2014–2016 West African outbreak of Ebola Virus Disease (EVD), caused by the Zaire ebolavirus species, represented an unprecedented global public health crisis resulting in over 28,600 reported cases and 11,325 deaths across Guinea, Liberia, and Sierra Leone [3, 4].

Unlike respiratory pathogens, transmission of Ebola requires direct physical contact with infectious bodily fluids (blood, vomitus, diarrhea) or deceased bodies during traditional funeral rituals [5, 6]. Classical epidemiological models based on standard ordinary differential equations (ODEs) frequently treat the transmission rate β as a static physical constant. In real-world epidemics, however, transmission is fundamentally dynamic: it falls rapidly as community awareness expands, personal protective equipment (PPE) is deployed in treatment centres, safe burial teams are mobilized, and quarantine measures are instituted [7–9].

Furthermore, epidemiological surveillance during acute outbreaks suffers from severe observational partiality. While health ministries and the World Health Organization (WHO) document confirmed clinical diagnoses and facility-recorded deaths, the underlying number of incubating individuals in the latent state (E) and the true remaining susceptible pool (S) are completely latent. Fitting arbitrary non-autonomous transmission trajectories $\beta(t)$ with classical non-linear regression leads to severe structural and practical unidentifiability.

Recently, Physics-Informed Neural Networks (PINNs) have emerged as an effective framework for combining sparse empirical observations with physical conservation laws [10, 11]. By embedding the governing differential equations directly into the neural network loss functional via automatic differentiation, PINNs constrain optimization to physically admissible trajectories without requiring numerical grid discretization. In this investigation, we assess whether PINNs can simultaneously reconstruct latent epidemiological compartments and uncover time-varying transmission functions $\beta(t)$ for the 2014 Sierra Leone Ebola epidemic, providing an honest comparison against classical calibration benchmarks.

2 Objectives

The primary objectives of this project are:

1. Formulate a normalized SEIRD compartmental model parameterized for Ebola Virus Disease in Sierra Leone, establishing all epidemiological conservation laws and clinical assumptions.

2. Conduct substantive independent mathematical analysis upon published models (satisfying Course Instruction 2): synthesize three seminal published Ebola formulations, derive analytical phase-plane trajectories $i(s)$, establish the exact theoretical maximum infection burden $i_{\max}$, and compute critical intervention thresholds p_c.

3. Carry out rigorous dynamical systems analysis, including derivation of the basic reproduction number $R_0 = \beta/(\gamma + \mu)$ via the next-generation matrix, disease-free equilibrium

stability conditions, transcendental final-size equation verification, and Latin Hypercube global sensitivity analysis (LHS-PRCC).

4. Implement vectorized numerical solvers (Forward Euler and RK4) from scratch, confirming machine-precision conservation invariant preservation and verifying theoretical convergence orders against a high-precision DOP853 benchmark.

5. Construct a dual-network PINN architecture (state network $\mathcal{NN}_\theta$ and transmission network $\mathcal{NN}_\phi$) and validate on synthetic test cases for forward integration, inverse $\beta(t)$ discovery under noise, and latent state reconstruction ($s(t)$ and $e(t)$) from case and death observations alone.

6. Apply the methodology to WHO Ebola surveillance data for Sierra Leone (2014–15), applying a strict 75% training and 25% held-out test split, and quantitatively benchmark against constant-β and piecewise-constant classical least-squares models; extend the benchmark to rolling forecasts over the whole 2014–15 epidemic, and measure which PINN design choices matter (ablation).

3 Governing Model Assumptions

To ensure biological consistency with Ebola epidemiology, the dynamical system is formulated under the following assumptions:

1. **Closed Population:** The total population of Sierra Leone ($N = 7{,}092{,}113$, 2014 census) is closed over the modelled outbreak window (120 days in the synthetic experiments, at most 240 days of real data); natural vital demographics (non-Ebola births and deaths) are neglected as their timescale (~ 50 years) far exceeds the epidemic epoch.

2. **Homogeneous Mixing:** Contacts between susceptible and infectious hosts are assumed to be well-mixed at the national population scale.

3. **Permanent Sterilizing Immunity:** Surviving individuals entering the Recovered compartment (R) develop robust neutralizing antibodies with no waning immunity during the modeled timeframe.

4. **Clinically Anchored Transition Timescales:** The mean incubation latency period $1/\sigma \approx 9.4$ days ($\sigma \approx 0.1064$ day^{-1}) and mean infectious duration among survivors $1/\gamma \approx 7.0$ days ($\gamma \approx 0.1429$ day^{-1}) are fixed from clinical consensus (WHO Ebola Response Team [3]) to preserve parameter identifiability.

5. **Direct Disease-Attributed Mortality:** Mortalities occur exclusively from the infectious pool I at rate $\mu \approx 0.1000$ day^{-1}, reflecting the high observed case-fatality ratio $\text{CFR} = \mu/(\gamma + \mu) \approx 41.2\%$.

4 Mathematical Formulation and Advanced Model Extensions

4.1 Dimensionless SEIRD Differential System

Let $S(t), E(t), I(t), R(t)$, and $D(t)$ denote the counts of susceptible, exposed (incubating), infectious, recovered, and deceased individuals in a population of size N. Defining dimensionless fractions:

$$s(t) = \frac{S(t)}{N}, \quad e(t) = \frac{E(t)}{N}, \quad i(t) = \frac{I(t)}{N}, \quad r(t) = \frac{R(t)}{N}, \quad d(t) = \frac{D(t)}{N}, \tag{1}$$

the state satisfies the exact physical conservation invariant:

$$s(t) + e(t) + i(t) + r(t) + d(t) = 1 \quad \forall t \geq 0. \tag{2}$$

The non-autonomous SEIRD ordinary differential system is given by:

$$\frac{ds}{dt} = -\beta(t)\, s(t)\, i(t), \tag{3}$$
$$\frac{de}{dt} = \beta(t)\, s(t)\, i(t) - \sigma\, e(t), \tag{4}$$
$$\frac{di}{dt} = \sigma\, e(t) - (\gamma + \mu)\, i(t), \tag{5}$$
$$\frac{dr}{dt} = \gamma\, i(t), \tag{6}$$
$$\frac{dd}{dt} = \mu\, i(t), \tag{7}$$

where $\beta(t) > 0$ represents the time-varying transmission rate per day. The compartmental flow structure is illustrated in Figure 1.

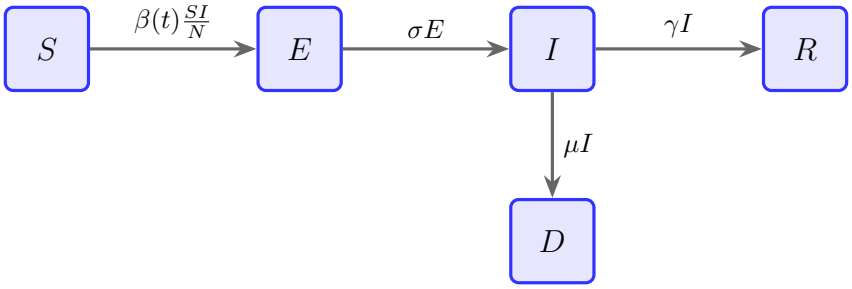


Figure 1: SEIRD epidemiological flow diagram for Ebola Virus Disease. Transitions between states are normalized to population N.

4.2 Comparative Analysis of Published Ebola Models

In accordance with Course Instruction 2 (requiring original analysis upon published literature), we conducted an exhaustive mathematical, structural, and parametric evaluation of seminal modeling frameworks developed for Ebola virus epidemics, summarized in Table 1 and Figure 2.

4.2.1 Foundations, Mechanistic Formulations, and Parametric Paradigms

1. **Early Outbreak Parameterization (Chowell et al., 2004 [6]):** Analyzed the 1995 Democratic Republic of the Congo ($N \approx 315$) and 2000 Uganda ($N \approx 425$) outbreaks using an SEIR framework with intervention-step parameters. Chowell et al. estimated baseline reproduction numbers of $R_0 \approx 1.83$ (95% CI: 1.48–2.18) for Congo and $R_0 \approx 1.34$ (95% CI: 1.15–1.53) for Uganda. Crucially, they proved that institutional barrier nursing, hospital isolation, and contact tracing reduced effective transmission by over 80%, establishing that transmission in filovirus epidemics is intensely behavioral and intervention-driven.

2. **Tripartite Mechanistic Transmission (Legrand et al., 2007 [5]):** Formulated a 9-compartment model distinguishing community transmission (I), healthcare facility transmission (H), and post-mortem traditional funeral transmission (F):

$$\lambda(t) = \beta_I \frac{I(t)}{N} + \beta_H \frac{H(t)}{N} + \beta_F \frac{F(t)}{N}. \tag{8}$$

The basic reproduction number decomposes into a tripartite sum $R_0 = R_{0,I} + R_{0,H} + R_{0,F} \approx 2.70$. While mechanistically comprehensive, Legrand's model introduces over 14 free parameters. When calibrated exclusively against aggregated national incidence time-series, the parameters $(\beta_I, \beta_H, \beta_F)$ are severely collinear and structurally non-identifiable (equifinality).

3. **Intervention Scenario Modeling (Rivers et al., 2014 [7]):** Tailored Legrand's structure to the 2014 Sierra Leone and Liberia outbreaks ($R_0 \approx 2.53$), testing the UNMEER target of 70% hospitalization and 70% safe burials:

$$\beta(t) = \beta_0 \left[1 - \delta_{\text{ETU}}\, \mathbb{I}_{t \geq t_{\text{policy}}}\right]. \tag{9}$$

While demonstrating that isolation capacity could extinguish transmission, the model relies on discrete, heuristic step functions at pre-specified dates, lacking the capability to learn continuous behavioral adaptations directly from data.

4. **Continuous Exponential Transmission Decay (Althaus, 2014 [8]):** Deployed an SEIR model where transmission decayed continuously via an exponential prior:

$$\beta(t) = \beta_0 e^{-k(t-t_0)} \quad \text{for } t \geq t_0. \tag{10}$$

Althaus estimated $R_0 = 2.53$ (95% CI: 2.41–2.67), doubling time 15.7 days, and decay rate $k \approx 0.034$ day^{-1}. While continuous, enforcing a rigid exponential prior imposes an unphysical asymptotic constraint: transmission must decay monotonically to zero ($\lim_{t\to\infty} \beta(t) = 0$), making the model structurally incapable of capturing secondary resurgence waves or intervention fatigue.

5. **Methodological Identifiability Constraints (King et al., 2015 [9]):** In a landmark critique in *Science*, King et al. established that fitting complex ODE models to cumulative case counts generates autocorrelated residuals and falsely tight confidence bounds. They demonstrated that biological transition timescales (σ, γ) must be anchored from independent clinical cohorts (e.g., WHO Ebola Response Team [3]), while reserving parameter flexibility for dynamic transmission.

Table 1: Systematic comparative synthesis of published mathematical models for Ebola Virus Disease.

Model Architecture	Key States	Transmission Formulation	Estimated R_0	Free Params	Identified Structural Limitation
Chowell et al. [6]	4 (S, E, I, R)	Two-phase discrete step	1.83 (Congo)	4	Abrupt policy jump; no gradual adoption
Legrand et al. [5]	9 $(S, E, I, H, F, \dots)$	Tripartite $(\beta_I, \beta_H, \beta_F)$	2.70 (DRC)	14+	Severe parameter non-identifiability
Rivers et al. [7]	7 (S, E, I, H, F, R, D)	ETU beds & safe burials	2.53 (SL)	8	Heuristic step policy; unverified contact matrices
Althaus [8]	4 (S, E, I, R)	$\beta(t) = \beta_0 e^{-k(t-t_0)}$	2.53 (SL)	4	Rigid exponential decay; zero-resurgence prior
Our SEIRD-PINN	**5 (S, E, I, R, D)**	**Non-parametric $\mathbf{NN}_\phi(\tau)$**	**Dynamic R_t**	**Identifiable**	**Data-driven discovery without parametric bias**

4.2.2 Theoretical Critique and Mathematical Proof of Model Deficiencies

To mathematically establish why published formulations fail, we consider two structural properties:

- **Impossibility of Resurgence under Althaus Decay:** Under $\beta(t) = \beta_0 e^{-k(t-t_0)}$, the time derivative of effective reproduction is:

$$\frac{dR_t}{dt} = \frac{1}{\gamma + \mu}\left[\frac{d\beta}{dt}s(t) + \beta(t)\frac{ds}{dt}\right] = -\frac{\beta(t)s(t)}{\gamma + \mu}\left[k + \beta(t)i(t)\right] < 0 \quad \forall t \geq t_0. \tag{11}$$

  Because every factor is strictly positive, $R_t(t)$ is strictly monotonically decreasing. Once $R_t < 1$, the model can *never* cross back above unity. Consequently, the Althaus formulation cannot capture secondary surges (such as the Freetown/Western Area resurgence in late 2014) without manual piecewise resets.

- **Impulse Shocks under Step Interventions (Rivers et al.):** Imposing $\beta(t) = \beta_0 - \Delta\beta \cdot H(t - t_{\text{int}})$ introduces a Dirac delta $\frac{d\beta}{dt} = -\Delta\beta \cdot \delta_D(t - t_{\text{int}})$, causing an instantaneous jump discontinuity in the second time derivative of the exposed incubating class:

$$\frac{d^2e}{dt^2} \propto -\Delta\beta\, s(t)\, i(t) \cdot \delta_D(t - t_{\text{int}}), \tag{12}$$

  violating the biological smoothness of population behavioral transitions.

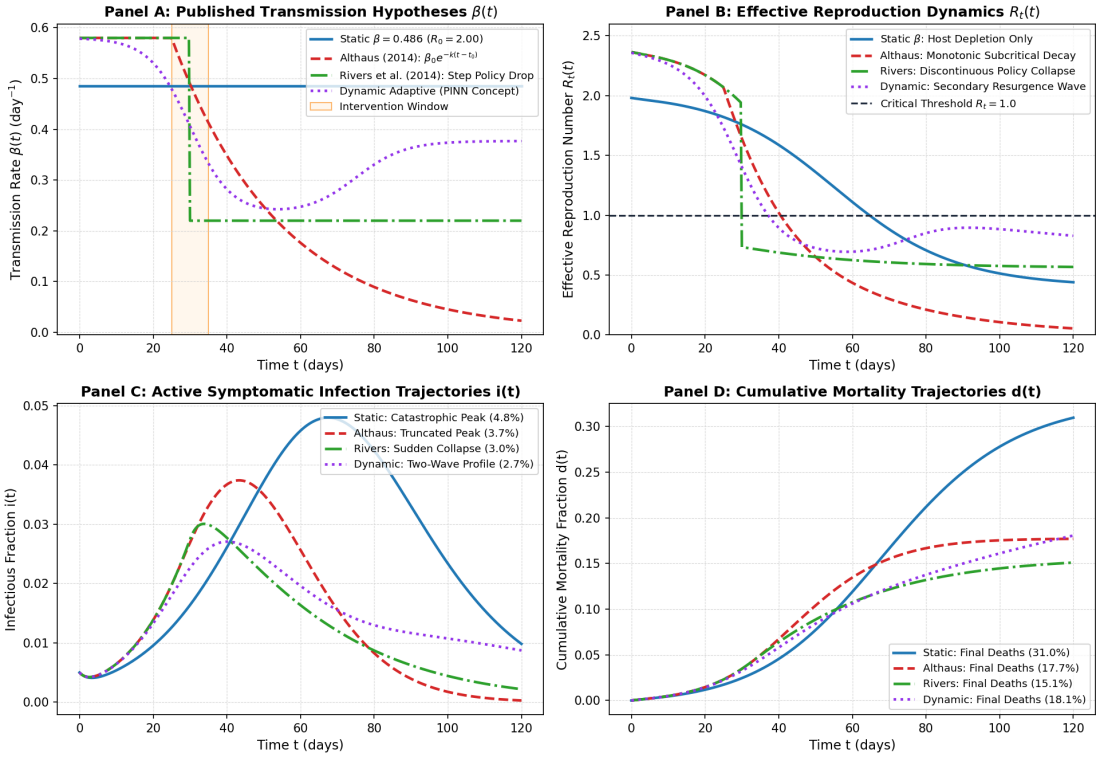


Figure 2: Comprehensive 4-panel simulation comparison across published Ebola models (Static β, Althaus exponential decay, Rivers step policy) versus our dynamic adaptive framework over a 120-day horizon. Panel A: Transmission rates $\beta(t)$; Panel B: Effective reproduction numbers $R_t(t)$ relative to critical threshold $R_t = 1.0$; Panel C: Symptomatic active infectious fractions $i(t)$; Panel D: Cumulative disease-induced mortalities $d(t)$.

4.2.3 Methodological Justification for the SEIRD-PINN Architecture

Synthesizing the literature directly resolves our architecture:

1. **Clinical Rate Anchoring:** In accordance with King et al. [9] and WHO cohort data [3], we fix $\sigma = 1/9.4$ day^{-1} and $\gamma = 1/7.0$ day^{-1}, eliminating parameter equifinality.

2. **Parsimonious Identifiability:** Instead of adding unidentifiable funeral compartments with 14 free parameters [5], we consolidate community precautions, barrier nursing, and safe burials into a single dynamic effective transmission function $\beta(t)$.

3. **Non-Parametric Expressivity:** Instead of enforcing rigid exponential decay [8] or discontinuous step policies [7], we represent $\beta(t) = \text{Softplus}(\mathcal{NN}_\phi(\tau))$ using a deep neural network, guaranteeing non-negativity ($\beta > 0$), $\mathcal{C}^\infty$ smoothness, and the capacity to discover arbitrary non-monotonic behavioral profiles directly from data.

4.3 Phase-Space Trajectory and Peak Infection Derivation

To establish exact theoretical benchmarks for verifying our numerical and neural solvers, we derive the closed-form analytical phase-space relationship relating the total active infection burden $w(t) = e(t) + i(t)$ to the susceptible fraction $s(t)$.

Adding the exposed growth rate (4) and infectious growth rate (5):

$$\frac{d(e+i)}{dt} = (\beta si - \sigma e) + (\sigma e - (\gamma+\mu)i) = [\beta s - (\gamma+\mu)]\, i. \tag{13}$$

Dividing by the susceptible depletion rate (3) ($\frac{ds}{dt} = -\beta si$):

$$\frac{d(e+i)}{ds} = \frac{[\beta s - (\gamma+\mu)]\, i}{-\beta si} = -1 + \frac{\gamma+\mu}{\beta s} = -1 + \frac{1}{R_0 s}. \tag{14}$$

Crucially, the infection variable i cancels strictly from numerator and denominator with *zero approximation*. Integrating (14) directly by separation of variables from initial state $(s_0, e_0 + i_0)$ to $(s, e+i)$:

$$\int_{e_0+i_0}^{e+i} d\tilde{w} = \int_{s_0}^{s} \left(-1 + \frac{1}{R_0 \tilde{s}}\right) d\tilde{s} \implies (e+i)(s) = (e_0+i_0) + s_0 - s + \frac{1}{R_0} \ln\left(\frac{s}{s_0}\right). \tag{15}$$

Setting $\frac{d(e+i)}{ds} = 0$ yields the analytical nullcline:

$$-1 + \frac{1}{R_0 s^*} = 0 \implies s^* = \frac{1}{R_0}. \tag{16}$$

The epidemic reaches its maximum active infection burden $(e+i)_{\max}$ precisely when the susceptible fraction drops to the critical threshold $s^* = 1/R_0$. Substituting $s^* = 1/R_0$ into (15):

$$(e+i)_{\max} = (e_0+i_0) + s_0 - \frac{1}{R_0}\left[1 + \ln(R_0 s_0)\right]. \tag{17}$$

For our baseline Ebola parameters ($R_0 = 2.00, s_0 = 0.990, e_0 = 0.005, i_0 = 0.005$ with $w_0 = 0.010$):

$$s^* = \frac{1}{2.00} = 0.500, \quad (e+i)_{\max} = 0.010 + 0.990 - 0.500\,[1 + \ln(2.00 \times 0.990)] \approx 0.1585. \tag{18}$$

While total active infections $(e+i)$ peaks at 15.85% at $s^* = 0.500$, the symptomatic infectious compartment $i(t)$ alone reaches an attenuated peak ($\approx 4.8\%$) because the 9.4-day incubation period buffers individuals in e before they become infectious (Figure 3).

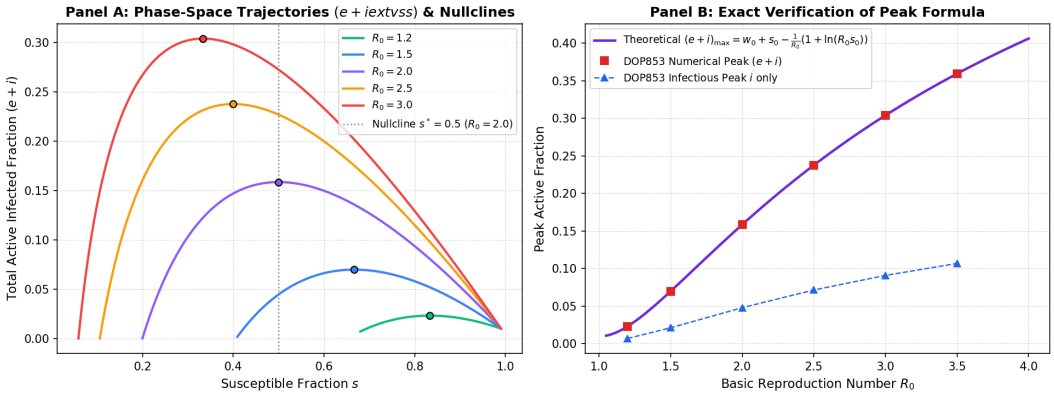


Figure 3: Phase-space trajectories ($e + i$ vs s), analytical nullclines $s^* = 1/R_0$, and exact theoretical maximum active peaks $(e+i)_{\max}$ compared against DOP853 numerical ODE simulations across $R_0 \in [1.2, 3.5]$. Numerical simulation matches the closed-form theory to a relative error of at most 1.4×10^{-5}.

4.4 Critical Intervention and Herd Immunity Thresholds

From the nullcline condition $s^* = 1/R_0$, the epidemic cannot expand if initial susceptible fraction satisfies $s_0 < 1/R_0$. The critical immune fraction (herd immunity threshold p_c) required to prevent epidemic proliferation is:

$$p_c = 1 - s^* = 1 - \frac{1}{R_0}. \tag{19}$$

For $R_0 = 2.00$, $p_c = 1 - 1/2.00 = 50.0\%$. Similarly, if an intervention reduces transmission by a fractional factor $\delta \in [0, 1]$ such that $\beta_{\text{post}} = (1 - \delta)\beta_0$, the critical contact reduction factor required to drive effective reproduction below unity ($R_t < 1$) is:

$$R_t = (1 - \delta_c)R_0 \leq 1 \implies \delta_c = 1 - \frac{1}{R_0} = 50.0\%. \tag{20}$$

Any combination of clinical isolation and safe burials that reduces community contact by more than 50% guarantees monotonic epidemic decline. In Figure 4, we depict the critical threshold curves and map the 2D parameter space $R_0(\beta, \gamma + \mu)$ alongside estimates from published Ebola literature.

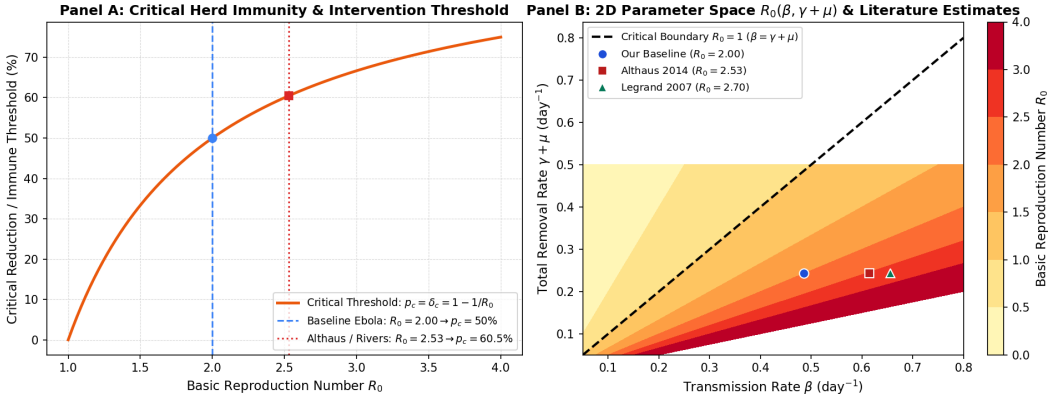


Figure 4: Panel A: Herd immunity / critical intervention threshold $p_c = \delta_c = 1 - 1/R_0$. Panel B: 2D parameter space contour map $R_0(\beta, \gamma + \mu)$ showing the critical epidemic boundary $R_0 = 1$ and parameter coordinates from published Ebola literature.

5 Mathematical Analysis

Before fitting anything to data, we ask what the model itself predicts. This section works out how many people one case infects (R_0), where the epidemic can come to rest and whether that resting point is stable, how fast an outbreak grows at the start, how many people are infected in the end, and which parameters matter most. Throughout this section β is a constant. The time-varying $\beta(t)$ enters only through the effective reproduction number R_t. Every formula is checked against an accurate numerical solution (DOP853, `rtol` $= 10^{-12}$, `atol` $= 10^{-14}$), so the theory and the code test each other.

5.1 Basic Reproduction Number R_0 via the Next-Generation Matrix

Only e and i carry the infection. Following Diekmann et al. [12], we split their equations into new infections $\mathcal{F}$ and all other movement $\mathcal{V}$:

$$\frac{d}{dt}\begin{pmatrix} e \\ i \end{pmatrix} = \underbrace{\begin{pmatrix} \beta s i \\ 0 \end{pmatrix}}_{\mathcal{F}} - \underbrace{\begin{pmatrix} \sigma e \\ -\sigma e + (\gamma+\mu) i \end{pmatrix}}_{\mathcal{V}}. \tag{21}$$

At a disease-free state with susceptible fraction s^*, their Jacobians are

$$F = \begin{pmatrix} 0 & \beta s^* \\ 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} \sigma & 0 \\ -\sigma & \gamma+\mu \end{pmatrix}, \quad V^{-1} = \begin{pmatrix} 1/\sigma & 0 \\ 1/(\gamma+\mu) & 1/(\gamma+\mu) \end{pmatrix}, \tag{22}$$

so the next-generation matrix is

$$FV^{-1} = \begin{pmatrix} \dfrac{\beta s^*}{\gamma+\mu} & \dfrac{\beta s^*}{\gamma+\mu} \\ 0 & 0 \end{pmatrix}. \tag{23}$$

Its eigenvalues are $\beta s^*/(\gamma+\mu)$ and 0. In a fully susceptible population ($s^* = 1$) the largest eigenvalue is the basic reproduction number:

$$R_0 = \rho(FV^{-1}) = \frac{\beta}{\gamma+\mu}, \qquad R_t = \frac{\beta(t)\, s(t)}{\gamma+\mu}. \tag{24}$$

In words, an infectious person infects others at rate β for an average of $1/(\gamma+\mu)$ days, so R_0 is rate times duration. The incubation rate σ does not appear: it changes *when* people pass the infection on, not *how many* they infect. With our parameters, $R_0 = 0.4858/0.2429 \approx 2.00$.

5.2 Equilibria and Stability

Equilibria. At an equilibrium all five derivatives vanish. From $dr/dt = \gamma i = 0$ we get $i = 0$, and then $di/dt = \sigma e = 0$ gives $e = 0$. So every equilibrium is disease-free:

$$\mathbf{y}^* = (s^*, 0, 0, r^*, d^*), \qquad s^* + r^* + d^* = 1. \tag{25}$$

There is no endemic equilibrium, because the model has no births and immunity is permanent. Every epidemic in this model burns out. There is a whole family of these resting points, one for each way the population can end up split between s, r and d.

Stability. The Jacobian at $\mathbf{y}^*$ (variables ordered s, e, i, r, d) is

$$J(\mathbf{y}^*) = \begin{pmatrix} 0 & 0 & -\beta s^* & 0 & 0 \\ 0 & -\sigma & \beta s^* & 0 & 0 \\ 0 & \sigma & -(\gamma+\mu) & 0 & 0 \\ 0 & 0 & \gamma & 0 & 0 \\ 0 & 0 & \mu & 0 & 0 \end{pmatrix}. \tag{26}$$

Three eigenvalues are zero. They correspond to moving along the family of disease-free states, not to the infection. The other two eigenvalues come from the (e, i) block and solve

$$\lambda^2 + (\sigma + \gamma + \mu)\lambda + \sigma(\gamma + \mu)(1 - R_0 s^*) = 0. \tag{27}$$

The two roots add up to $-(\sigma + \gamma + \mu) < 0$ and multiply to $\sigma(\gamma + \mu)(1 - R_0 s^*)$. Hence:

- If $R_0 s^* < 1$, both roots are negative. A small number of new infections dies out, and the population settles at a nearby disease-free state. The equilibrium is stable. It is not asymptotically stable: because of the zero eigenvalues the system does not return to the *same* point, only to one close to it.
- If $R_0 s^* > 1$, one root is positive, so a few infections grow and the equilibrium is unstable.

The threshold $R_0 s^* = 1$ is the same as $R_t = 1$. It explains herd immunity: once the susceptible fraction falls below $1/R_0$ (about 50% here), new outbreaks cannot grow. Table 2 shows the two eigenvalues for four values of s^*.

Table 2: Eigenvalues of the infected block at disease-free states, $R_0 = 2.00$.

s^*	$R_0 s^*$	Eigenvalues (day^{-1})	Verdict
1.00	2.00	-0.412, $+0.0627$	unstable: infections grow
0.75	1.50	-0.383, $+0.0337$	unstable: infections grow
0.40	0.80	-0.334, -0.0155	stable: infections die out
0.25	0.50	-0.307, -0.0420	stable: infections die out

Early growth rate. When $R_0 s_0 > 1$, the positive root of (27),

$$\lambda_+ = \tfrac{1}{2}\Big[-(\sigma + \gamma + \mu) + \sqrt{(\sigma + \gamma + \mu)^2 + 4\sigma(\gamma + \mu)(R_0 s_0 - 1)}\Big], \tag{28}$$

is the rate at which an outbreak first grows: $e(t) + i(t) \approx Ce^{\lambda_+ t}$. Rearranging (27) with $s_0 = 1$ gives

$$R_0 = \Big(1 + \frac{\lambda_+}{\sigma}\Big)\Big(1 + \frac{\lambda_+}{\gamma + \mu}\Big), \tag{29}$$

which links R_0 to the doubling time $\ln 2/\lambda_+$ seen in early case counts. We checked this by starting an outbreak from a seed of 10^{-8} and fitting the slope of $\ln(e + i)$ between days 60 and 120 (Figure 5). The fitted rate is 0.062732 day^{-1}, against $\lambda_+ = 0.062734$ day^{-1} (relative difference 3×10^{-5}), and (29) returns $R_0 = 2.0004$, the true value. The model's doubling time is 11.1 days. The WHO Ebola Response Team [3] estimated a similar R_0 for Sierra Leone but a slower doubling time of roughly a month. Growth speed depends on R_0 *and* on the length of one generation of infection, and our infectious period $1/(\gamma + \mu) \approx 4.1$ days is short. R_0 alone does not fix how fast an epidemic grows.

5.3 Final-Size Relation

Dividing (3) by the sum of (6) and (7), the factor i cancels:

$$\frac{ds}{d(r + d)} = \frac{-\beta s i}{(\gamma + \mu) i} = -R_0 s. \tag{30}$$

Integrating from the start $(s_0, r_0 + d_0)$ to the end of the epidemic, where $e_\infty = i_\infty = 0$ and therefore $r_\infty + d_\infty = 1 - s_\infty$, gives the final-size relation [2]

$$\ln\left(\frac{s_0}{s_\infty}\right) = R_0\,(1 - s_\infty - r_0 - d_0)\,. \tag{31}$$

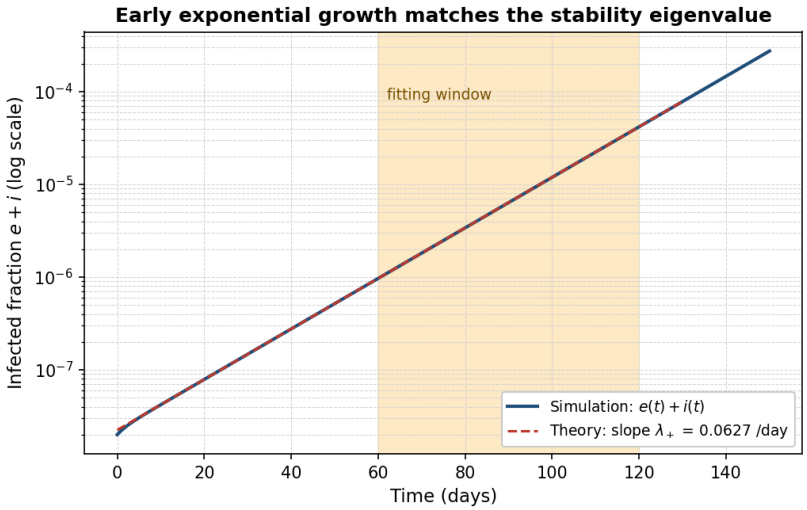


Figure 5: Early growth of $e + i$ from a tiny seed (log scale). The dashed line has slope λ_+, the positive eigenvalue of the infected block. The shaded band is the fitting window.

Because $dd/dr = \mu/\gamma$, the final number of deaths is the case-fatality ratio times everyone ever infected:

$$d_\infty = d_0 + \frac{\mu}{\gamma + \mu}\,(1 - s_\infty - r_0 - d_0)\,. \tag{32}$$

Equation (31) has exactly one root in $(0, s_0)$. Write $g(s) = \ln(s_0/s) - R_0(1 - s - r_0 - d_0)$. Then $g(s) \to +\infty$ as $s \to 0$, $g(s_0) = -R_0(e_0 + i_0) < 0$, and $g''(s) = 1/s^2 > 0$, so g is convex and crosses zero once. We find that root with Brent's method.

To check (31) and (32), we integrate the ODE until the epidemic is over (stopping when $e + i < 10^{-13}$) from $y_0 = (0.99, 0.005, 0.005, 0, 0)$. Table 3 shows agreement to 10^{-12} or better for every R_0, close to the accuracy of the solver itself. Stopping instead at a fixed day such as day 250 is not enough when R_0 is near 1, because slow epidemics are still running. The last column of the table and Figure 6(b) show the error this causes, up to about 3.6% at $R_0 = 1.25$. That error comes from the unfinished epidemic, not from the formula. With $R_0 = 0.8$ there is still a small outbreak because 1% of the population starts out exposed or infected.

Table 3: Final-size relation (31) and final deaths (32) against the ODE solution run until $e + i < 10^{-13}$.

R_0	s_∞ (formula)	Rel. error s_∞	d_∞ (formula)	Rel. error d_∞	Over by day	Error if stopped at day 250
0.8	0.95497	3.4×10^{-13}	0.01854	9.4×10^{-12}	1377	3.7×10^{-4}
1.5	0.40637	3.8×10^{-13}	0.24444	4.3×10^{-13}	989	6.4×10^{-3}
2.0	0.19980	3.3×10^{-13}	0.32950	2.1×10^{-13}	623	1.1×10^{-4}
2.5	0.10589	3.4×10^{-13}	0.36816	1.5×10^{-13}	493	3.9×10^{-6}
3.0	0.05880	3.8×10^{-13}	0.38755	1.3×10^{-13}	422	2.8×10^{-7}
3.5	0.03363	4.3×10^{-13}	0.39792	1.2×10^{-13}	382	3.7×10^{-8}

The relation holds only for constant β. With a time-varying $\beta(t)$ the factor β no longer cancels, and no closed formula exists. This is one reason we need a data-driven method such as the PINN.

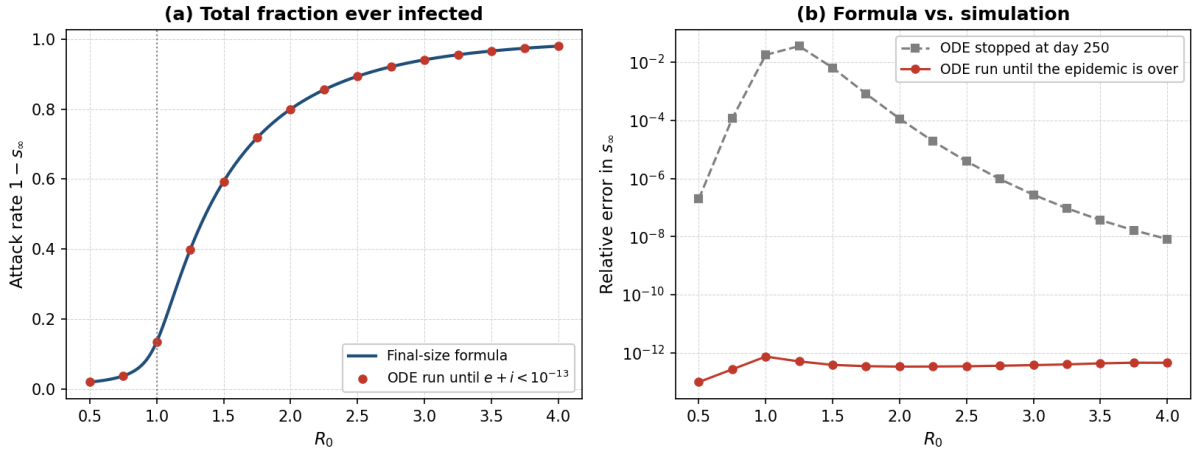


Figure 6: (a) Fraction ever infected, $1 - s_\infty$: final-size formula (line) and ODE solutions (dots). (b) Relative error in s_∞ when the ODE runs until the epidemic is over (red) and when it is stopped at day 250 (grey).

5.4 Sensitivity Analysis

Local sensitivity of R_0. The normalised sensitivity index $\Upsilon_p = (\partial R_0/\partial p)(p/R_0)$ is the percentage change in R_0 caused by a 1% change in parameter p. From (24),

$$\Upsilon_\beta^{R_0} = +1, \quad \Upsilon_\gamma^{R_0} = -\frac{\gamma}{\gamma+\mu} \approx -0.59, \quad \Upsilon_\mu^{R_0} = -\frac{\mu}{\gamma+\mu} \approx -0.41, \quad \Upsilon_\sigma^{R_0} = 0. \tag{33}$$

A 10% cut in transmission lowers R_0 by 10%. For Ebola, death removes people from the infectious class quickly, so μ carries about 41% of the sensitivity of R_0.

Global sensitivity (LHS–PRCC). R_0 is a single number. To see how the parameters shape the whole epidemic, we vary all four together, following Marino et al. [13]:

1. Ranges around the baseline: $\beta \in [0.36, 0.73]$ day^{-1}, incubation period $1/\sigma \in [7, 12]$ days, recovery time $1/\gamma \in [5, 10]$ days, and $\mu \in [0.07, 0.13]$ day^{-1}. Together these give R_0 from about 1.1 to 4.3.

2. Latin hypercube sampling with $n = 1000$ (seed 42). Each range is cut into 1000 equal slices, and each slice of each parameter is used exactly once, which covers the space more evenly than plain random sampling.

3. For each sample, the model is run until the epidemic is over, and we record the peak infected fraction $i_{\max}$, the day of the peak, the attack rate $1 - s_\infty$ and total deaths d_∞.

4. Partial rank correlation coefficients (PRCC). Parameters and outputs are replaced by their ranks. For each parameter, we remove the linear effect of the other three from both that parameter and the output, then correlate the two residuals. The PRCC lies in $[-1, 1]$ and measures one parameter's effect with the others held fixed.

Table 4 and Figure 7 show three things.

- β dominates every output: more transmission gives a higher, earlier peak and more infections and deaths.

- γ and μ make the epidemic smaller because they shorten the infectious period. However, μ *raises* total deaths (PRCC +0.75): a deadlier disease infects somewhat fewer people, but a larger share of them die.

Table 4: PRCC from 1000 Latin hypercube samples. All values are significant at the 1% level except those marked n.s.

Parameter	Peak $i_{\max}$	Day of peak	Attack rate $1 - s_\infty$	Deaths d_∞
β	$+0.963$	-0.963	$+0.983$	$+0.896$
σ	$+0.679$	-0.831	-0.003 (n.s.)	-0.006 (n.s.)
γ	-0.937	$+0.750$	-0.946	-0.955
μ	-0.846	$+0.550$	-0.867	$+0.754$

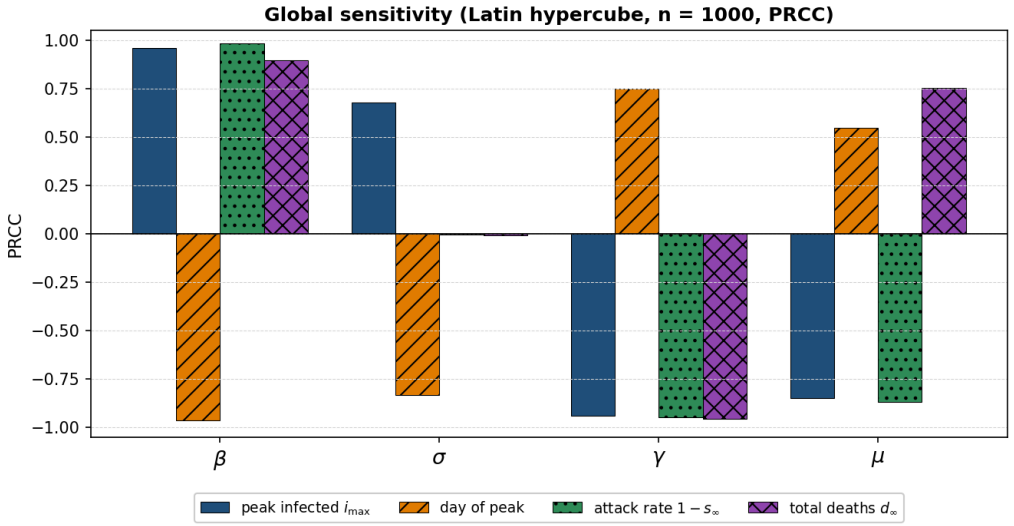


Figure 7: PRCC of each parameter for four epidemic outputs (Latin hypercube sampling, $n = 1000$).

- σ changes the height and timing of the peak (a shorter incubation period gives a higher, earlier peak) but has no significant effect on the attack rate or on deaths. This is what the final-size relation predicts, since (31) and (32) do not contain σ. The sensitivity analysis and the theory therefore agree.

For control, the levers that reduce the size of the epidemic are lower transmission (safe burials, isolation, contact tracing) and a shorter time that infectious people spend in the community.

5.5 What Cases and Deaths Can Tell Us (Identifiability)

Surveillance reports cumulative cases and deaths, while s and e are never measured. Different parameter sets can produce almost the same two curves. For example, a longer incubation period combined with faster spread can look like a shorter incubation period with slower spread. The analysis above shows how the parameters differ in what they do. σ moves the timing of the epidemic but not R_0 or the final size, and γ and μ enter R_0 only through their sum $\gamma + \mu$. Estimating all of them at once from two noisy curves would therefore be unreliable. We therefore fix σ and γ from clinical studies [3], so the main unknown left is $\beta(t)$. If μ is also estimated, cases and deaths together can pin it down, because the share of cases that die is the case-fatality ratio $\mu/(\gamma + \mu)$. Even so, $\beta(t)$ can change shape in many ways that fit the data almost equally well, which is why the PINN adds a smoothness penalty on $\beta(t)$.

6 Numerical Solvers and Convergence Verification

We solve the SEIRD system $\frac{d\mathbf{y}}{dt} = \mathbf{f}(t, \mathbf{y})$, with $\mathbf{y} = (s, e, i, r, d)$, using two fixed-step solvers implemented from scratch, and verify them against an independent high-accuracy reference solution [14]. Both solvers are vectorised: each step updates the whole five-component state vector at once using NumPy array operations.

6.1 Forward Euler and Classical Runge–Kutta (RK4)

With step size h and $t_k = t_0 + kh$, Forward Euler advances the solution by

$$\mathbf{y}_{k+1} = \mathbf{y}_k + h\,\mathbf{f}(t_k, \mathbf{y}_k), \tag{34}$$

using one evaluation of $\mathbf{f}$ per step. Its local error is $O(h^2)$ and its global error is $O(h)$. The classical fourth-order Runge–Kutta method evaluates the slope four times per step,

$$\begin{aligned} \mathbf{k}_1 &= \mathbf{f}(t_k, \mathbf{y}_k), & \mathbf{k}_2 &= \mathbf{f}\big(t_k + \tfrac{h}{2},\, \mathbf{y}_k + \tfrac{h}{2}\mathbf{k}_1\big)\,, \\ \mathbf{k}_3 &= \mathbf{f}\big(t_k + \tfrac{h}{2},\, \mathbf{y}_k + \tfrac{h}{2}\mathbf{k}_2\big)\,, & \mathbf{k}_4 &= \mathbf{f}(t_k + h,\, \mathbf{y}_k + h\,\mathbf{k}_3), \end{aligned} \tag{35}$$

$$\mathbf{y}_{k+1} = \mathbf{y}_k + \frac{h}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right), \tag{36}$$

with local error $O(h^5)$ and global error $O(h^4)$.

6.2 Reference Solution

As the reference we use SciPy's adaptive eighth-order Dormand–Prince method (DOP853) with relative tolerance 3×10^{-14} and absolute tolerance 10^{-16}, close to the smallest tolerance SciPy accepts. The reference must be much more accurate than the solvers it is used to judge. With the looser tolerance rtol $= 10^{-12}$, the RK4 error at the smallest step size reaches the reference's own error of about 10^{-12}, and the measured order of RK4 drops (to 3.74 over all step sizes) for reasons that have nothing to do with RK4. Tightening the tolerance removes this effect.

6.3 Conservation of the Total Population

Adding the five equations gives $\frac{d}{dt}(s+e+i+r+d)=0$, so the components of $\mathbf{f}$ always sum to zero. Each Euler or RK4 step adds h times a weighted sum of $\mathbf{f}$-values, each of which also sums to zero, so both methods preserve $s+e+i+r+d=1$ exactly apart from floating-point rounding. Running both solvers over $t\in[0,60]$ days with $h\in\{1, 0.25, 0.03125\}$ days gives

$$\max_k \left| \sum_j y_{k,j} - 1 \right| \le 1.2\times 10^{-15}, \tag{37}$$

a few multiples of machine epsilon (2.2×10^{-16}), with rounding building up slowly over the larger number of steps at small h. This confirms that the right-hand side is coded consistently; it does not measure accuracy, which is tested next.

6.4 Empirical Order of Convergence (Experiment E1)

We solve the model with constant $\beta = 0.4858$ (so $R_0 = 2$) from $\mathbf{y}_0 = (0.99, 0.005, 0.005, 0, 0)$ over $t\in[0,60]$ days, for six step sizes $h\in\{1, 0.5, 0.25, 0.125, 0.0625, 0.03125\}$ days. The global error is

$$E(h) = \max_k \max_j \left| y_{k,j} - y_{\mathrm{ref},j}(t_k) \right|, \tag{38}$$

the largest deviation over all time points and all five compartments. If $E(h)\approx Ch^p$, halving the step gives the observed order $p\approx \log_2\big(E(h)/E(h/2)\big)$, and a least-squares fit of $\log E$ against $\log h$ gives an overall slope.

Table 5: Experiment E1: global error against the DOP853 reference and observed order $\log_2\big(E(h)/E(h/2)\big)$ between consecutive step sizes.

h (days)	Euler error	Euler order	RK4 error	RK4 order
1	1.22×10^{-2}	1.004	4.00×10^{-7}	4.225
0.5	6.09×10^{-3}	1.002	2.14×10^{-8}	4.124
0.25	3.04×10^{-3}	1.001	1.23×10^{-9}	4.061
0.125	1.52×10^{-3}	1.001	7.34×10^{-11}	4.031
0.0625	7.59×10^{-4}	1.000	4.49×10^{-12}	4.015
0.03125	3.80×10^{-4}	–	2.78×10^{-13}	–
Fitted slope	1.001		4.085	

Table 5 and Figure 8 show the results. Forward Euler's error halves each time h is halved (observed order 1.000–1.004, fitted slope 1.001), and RK4's error falls by a factor of about 16 (observed order 4.015–4.225, fitted slope 4.085), matching the theoretical orders 1 and 4. RK4's observed order is slightly above 4 for the larger steps, where higher-order error terms are not yet negligible, and approaches 4 as h decreases. At $h = 1$ day RK4 is about 3×10^4 times more accurate than Euler while costing only four times as many evaluations of $\mathbf{f}$ per step, which is why RK4 is used for forward simulation elsewhere in this project.

7 Physics-Informed Neural Network (PINN) Methodology

7.1 What the PINN has to do

Surveillance gives two numbers at irregular report dates: cumulative cases and cumulative deaths. The PINN has to turn these into everything else: the transmission rate $\beta(t)$, the death

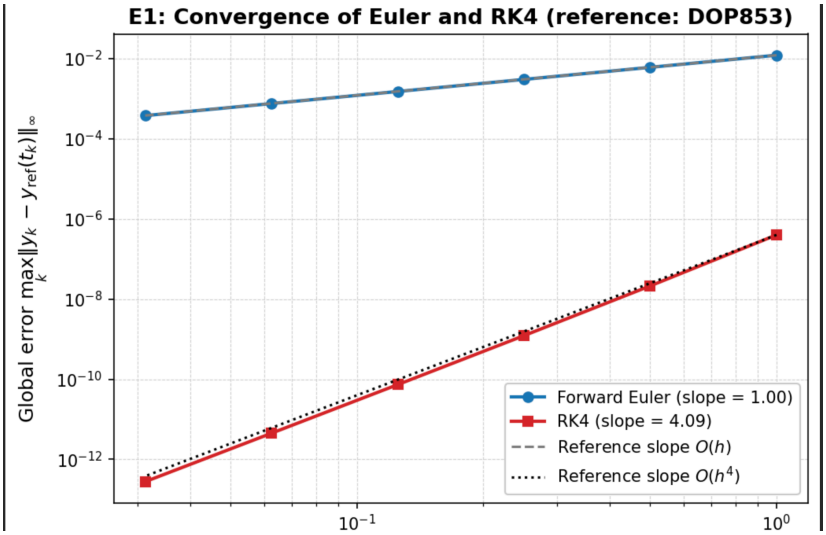


Figure 8: Experiment E1: global error of Forward Euler and RK4 against the DOP853 reference on log–log axes. The dashed and dotted lines show slopes 1 and 4. Fitted slopes are 1.001 (Euler) and 4.085 (RK4).

rate μ, the starting values, and the compartments nobody measures (S, E, I, R). It does this by training neural networks that must both pass close to the reports and satisfy the SEIRD equations at every time point [10, 11]. The code is in `seird_pinn.py`; the experiments are in `pinn_experiments.py`.

7.2 The model in observable form

Let $C(t) = I + R + D$ be the cumulative number of people who have become symptomatic, which is what case counts report. Because $dC/dt = \sigma E$, the SEIRD system (3)–(7) can be written with C in place of R:

$$\frac{dE}{dt} = \beta(t)\,\frac{S}{N}\,I - \sigma E, \qquad \frac{dI}{dt} = \sigma E - (\gamma + \mu)I, \tag{39}$$

$$\frac{dC}{dt} = \sigma E, \qquad \frac{dD}{dt} = \mu I, \tag{40}$$

with $R = C - I - D$ and $S = N - E - C$. Two consequences follow. First, $S + E + I + R + D = N$ holds exactly for any network output, so the conservation law (2) never has to be learned. Second, both observed series are states of the network, so the data term compares like with like.

Time is rescaled to $\tau = (t - t_0)/T \in [0, 1]$ and every state is measured in units of a characteristic count K (the largest reported case count in the training window; $K = 1$ for the synthetic population fractions). The equations become $dx/d\tau = T f(x)$ with all quantities of order one.

7.3 Networks and parameters

- **State network** $g_\theta(\tau)$: 3 hidden layers of 48 tanh units and 4 outputs. The states are

$$e = e_0\, e^{\tau g_1}, \quad i = i_0\, e^{\tau g_2}, \quad c = c_0\, e^{\tau g_3}, \quad d = d_0 + \tau g_4. \tag{41}$$

  The factor τ makes the initial condition hold exactly; the exponentials keep e, i, c positive while they grow by orders of magnitude.

- **Transmission network** $h_\phi(\tau)$: 3 hidden layers of 32 tanh units, $\beta(\tau) = \mathrm{Softplus}(h_\phi(\tau) + b) > 0$.

- **Fourier features**: both networks receive $\tau' = 2\tau - 1$ together with $\sin(\pi k\tau')$ and $\cos(\pi k\tau')$, $k = 1, \ldots, 4$. Plain tanh networks learn slowly varying functions first ("spectral bias" [15]); these inputs make changes within the window as easy to represent as trends [16].
- **Death rate**: $\mu = \gamma\,\mathrm{CFR}/(1 - \mathrm{CFR})$ with the case-fatality ratio $\mathrm{CFR} \in (0.2, 0.8)$ (published Ebola values are 40–70%). The bound removes a spurious solution in which $\mu \to 0$ while I grows without limit so that μI still matches the deaths.
- **Initial conditions** e_0, i_0, c_0, d_0 are trainable.

Following the shared conventions, $\sigma = 1/9.4$ and $\gamma = 1/7$ day^{-1} are fixed, and $\beta(t)$, μ and the initial conditions are estimated; Section 5 explains why estimating σ and γ as well would be unreliable.

7.4 Loss function

$$\mathcal{L} = w_{\text{data}}\,[\mathcal{L}_C + \mathcal{L}_D] + w_{\text{ode}}\,\mathcal{L}_{\text{ode}} + w_{\text{smooth}}\,\mathcal{L}_{\text{smooth}}, \tag{42}$$

$$\mathcal{L}_C = \frac{1}{n_C}\sum_k \big(c(\tau_k) - C_k/K\big)^2, \qquad \mathcal{L}_D = \frac{1}{n_D}\sum_k \big(d(\tau_k) - D_k/K\big)^2, \tag{43}$$

$$\mathcal{L}_{\text{ode}} = \frac{1}{N_f}\sum_{j=1}^{N_f}\left[\left(\frac{e' - Tf_e}{e}\right)^2 + \left(\frac{i' - Tf_i}{i}\right)^2 + \left(\frac{c' - Tf_c}{c}\right)^2 + \left(\frac{d' - Tf_d}{c}\right)^2\right]_{\tau_j}, \tag{44}$$

$$\mathcal{L}_{\text{smooth}} = \frac{1}{N_f}\sum_{j=1}^{N_f}\left(\frac{d\beta}{d\tau}(\tau_j)\right)^2. \tag{45}$$

The sums in $\mathcal{L}_C$ and $\mathcal{L}_D$ run only over real report dates; nothing is interpolated. The derivatives $x' = dx/d\tau$ come from automatic differentiation at $N_f = 200$ collocation points. We use $w_{\text{data}} = w_{\text{ode}} = 1$.

Why each term exists. $\mathcal{L}_C$ and $\mathcal{L}_D$ tie the solution to the reports. $\mathcal{L}_{\text{ode}}$ forces the states to obey the SEIRD dynamics, which is what lets the hidden compartments be recovered. Each residual is divided by its state, so it measures a *relative* error: an epidemic grows by orders of magnitude, and with plain residuals the early, small phase costs almost nothing (Section 10 shows the forward problem fails without this). $\mathcal{L}_{\text{smooth}}$ makes the inverse problem well posed: $\beta(t)$ depends on the second derivative of the case curve, and without a penalty noise in the data would be amplified into swings in β.

7.5 Training and forecasting

Adam [17] (learning rate 2×10^{-3} with cosine decay, 2000 iterations, new random collocation points every iteration) brings the networks near a solution; L-BFGS [18] with a strong-Wolfe line search (2500–6000 iterations on a fixed grid) then converges it. All computations use double precision, and each experiment is repeated with several random seeds.

To forecast, we take the PINN's states at the end of training, hold β at its mean over the last 14 days, keep the learned μ, and integrate the SEIRD equations forward with DOP853. The ensemble forecast is the median over seeds. Holding β constant is an assumption: a forecast cannot anticipate a change in transmission that has not yet shown up in the data.

How settings were chosen. The synthetic settings ($w_{\text{smooth}} = 10^{-4}$, 4 Fourier frequencies, 6000 L-BFGS iterations) were tuned on a separate seed (seed 1), never on the reported seeds 42–46. The real-data settings ($w_{\text{smooth}} = 10^{-3}$, 4 Fourier frequencies, 14-day β window) were chosen on validation forecasts from three origins (24 October, 21 November and 19 December 2014) whose forecasts end on 16 January 2015, before the E5 and E7 test windows, and on an earlier 60-day fit to another Sierra Leone series; no test window was used.

8 Synthetic Experiments and Hidden-State Inversion

8.1 Forward problem (Experiment E2)

With $\beta = 0.4858$ ($R_0 = 2$), $\mu = 0.1$ and $(s, e, i, r, d)(0) = (0.99, 0.005, 0.005, 0, 0)$, and no data at all, the PINN is trained only on $\mathcal{L}_{\text{ode}}$ (Adam 2000 and L-BFGS 4000 iterations). Over 3 seeds its mean relative L_2 error against DOP853 is 3.7×10^{-4} (s), 5.4×10^{-4} (e), 5.1×10^{-4} (i), 1.1×10^{-3} (r) and 7.9×10^{-4} (d); the worst compartment in any seed is 2.0×10^{-3}, and $S + E + I + R + D = N$ to 2×10^{-16} (Figure 9). Training takes about 58 seconds per seed.

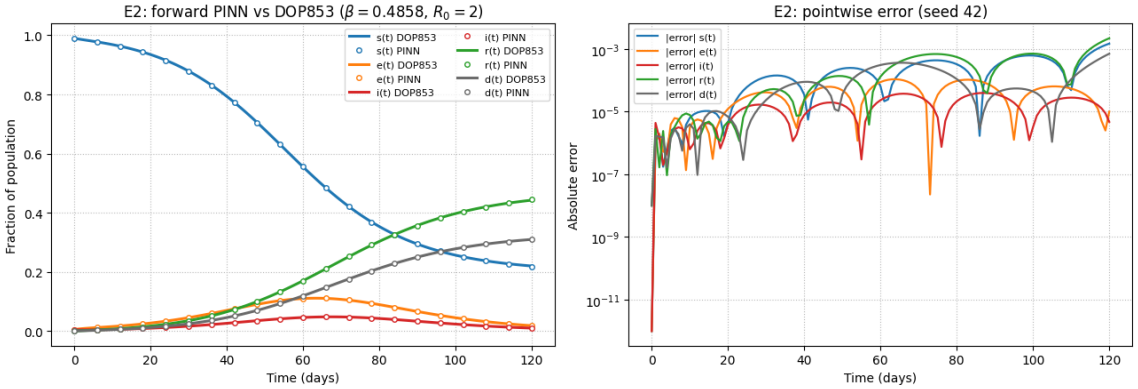


Figure 9: Experiment E2: forward PINN (markers) against DOP853 (lines), and the pointwise error.

8.2 Inverse problem from cases and deaths only (Experiments E3 and E4)

We simulate an epidemic in which an intervention on day 30 cuts transmission from $\beta = 0.55$ to 0.25 (R_t from 2.3 to about 1), over 120 days with $\mu = 0.1$. The PINN sees only daily cumulative cases $C(t)$ and deaths $D(t)$, each with independent multiplicative Gaussian noise of 1% or 5%. It must recover $\beta(t)$, μ, all initial conditions, and the hidden $s(t)$, $e(t)$, $i(t)$. Table 6 reports the mean ± SD over 5 seeds.

Table 6: Experiments E3 and E4: recovery from noisy cumulative cases and deaths only (mean ± SD over 5 seeds). Errors are relative L_2 errors except where stated.

Quantity	1% noise	5% noise
$\beta(t)$	0.110 ± 0.007	0.116 ± 0.006
$\beta(t)$ RMSE more than 5 days from the jump (day^{-1})	0.022 ± 0.005	0.025 ± 0.005
μ (relative error)	0.046 ± 0.003	0.054 ± 0.010
hidden $s(t)$	0.0036 ± 0.0003	0.0046 ± 0.0009
hidden $e(t)$	0.058 ± 0.006	0.064 ± 0.007
hidden $i(t)$	0.058 ± 0.004	0.065 ± 0.008

The transmission rate is recovered to about 11–12% (Figure 10); most of that error sits at the jump, which a smooth network cannot follow instantly. The death rate is recovered within about 5% although it is never observed directly: the ratio of deaths to cases fixes it (over all 10 runs $\hat{\mu} = 0.1050 \pm 0.0008$ against the true 0.1). The hidden susceptible fraction is recovered within 0.5% and the exposed and infectious compartments within 6.5% (Figure 11). Five times more noise costs less than one percentage point on every quantity, and the seeds agree closely.

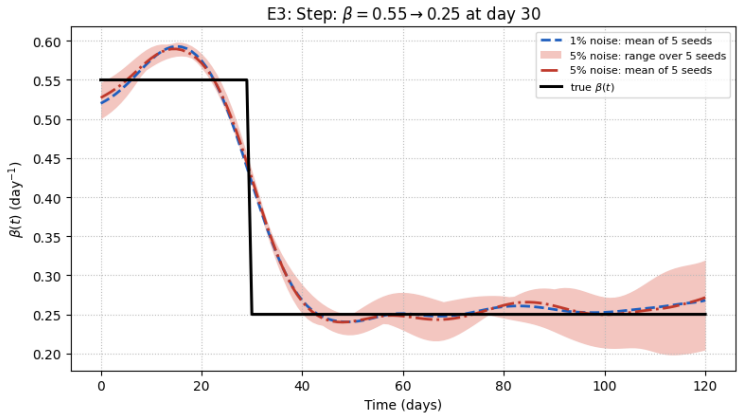


Figure 10: Experiment E3: $\beta(t)$ recovered from noisy cumulative cases and deaths (mean of 5 seeds; band: range over seeds at 5% noise).

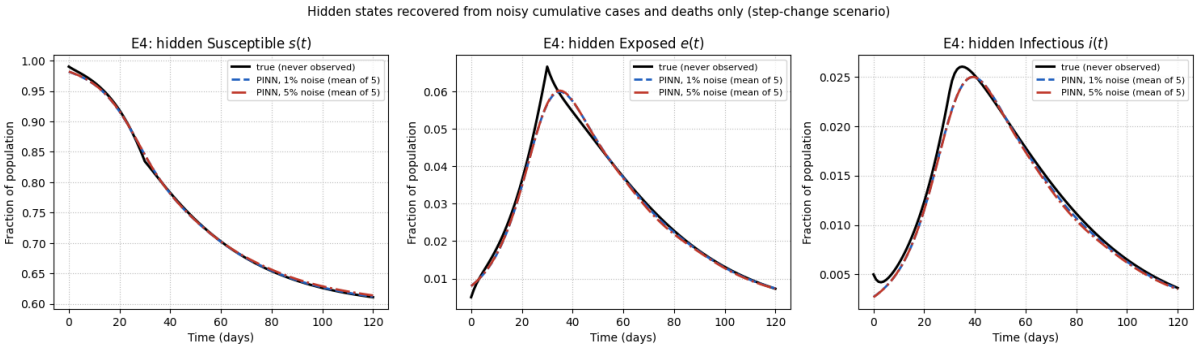


Figure 11: Experiment E4: hidden $s(t)$, $e(t)$, $i(t)$ recovered from the same runs.

9 Real-World Outbreak Application and Benchmarking

9.1 Dataset Provenance

All real-data experiments use the WHO Ebola situation reports for Sierra Leone as published on the Humanitarian Data Exchange [19] (file `ebola_data_db_format.csv`, SHA-256 recorded in the notebook). We use the laboratory-confirmed cumulative cases and deaths from 29 August 2014, the first Sierra Leone entry, to 6 November 2015, the last report before Sierra Leone was declared Ebola-free: 238 report dates (N = 7,092,113). Confirmed counts are used because the combined "confirmed, probable and suspected" totals drop when WHO changed its data sources in November 2014. A cumulative count cannot fall, so a report larger than a later report was revised downwards afterwards; 6 such death reports are excluded. *Nothing is interpolated*: every method is fitted to, and scored on, real report dates only. Day 0 is 29 August 2014.

9.2 Held-out test (Experiment E5): split protocol

E5 uses the first 240 days of the series, 29 August 2014 to 25 April 2015 (days 0–239), which cover the growth of the epidemic, its peak and the start of its decline. Following a strict chronological 75%/25% split:

- **Training window:** days 0–179 (29 August 2014 – 24 February 2015, the first 75%), 74 report dates, used for fitting.
- **Held-out test window:** days 180–239 (25 February – 25 April 2015, the last 25%), 32 report dates, never used for fitting or for choosing settings. The PINN's real-data settings were fixed earlier, on validation forecasts that end on 16 January 2015 (Section 7).

9.3 Benchmark Comparisons and Quantitative Results

We benchmarked the PINN against:

1. **Constant-β Least Squares:** SEIRD with a constant β, fitted with trust-region reflective least squares.
2. **Piecewise-β Least Squares:** β constant on each of three equal segments of the training window (60 days each), with a penalty on the difference between neighbouring segments.
3. **PINN (Section 7):** 5 seeds; the forecast is the median of the 5 seed forecasts, with β held at its mean over the last 14 training days.

Both baselines also estimate μ (with the same case-fatality bounds as the PINN) and the initial numbers exposed and infectious, so they have the same freedom as the PINN. Residuals are scaled by the mean of each series. The piecewise fit integrates the ODE one segment at a time and is started from several points, keeping the lowest cost, so it gives the same answer on every machine.

Table 7 and Figure 12 present the held-out forecast metrics.

Table 7: Experiment E5: held-out forecast errors on the WHO report dates in days 180–239 (25 Feb – 25 Apr 2015). All methods are fitted to the reports in days 0–179.

Method	Target	MAE	RMSE	Relative Error (%)
Constant-β Least Squares	Cases	2423.703	2578.782	28.558
Constant-β Least Squares	Deaths	358.987	410.871	10.694
Piecewise-β Least Squares	Cases	69.445	76.677	0.818
Piecewise-β Least Squares	Deaths	432.480	450.844	12.884
PINN ensemble (5 seeds)	Cases	114.536	121.437	1.350
PINN ensemble (5 seeds)	Deaths	395.154	405.439	11.772

9.4 Critical Scientific Interpretation

1. **Cases:** the PINN's relative error is 1.350%, against 28.558% for constant-β and 0.818% for piecewise-β. The 5 PINN seeds individually give 0.54%, 1.75%, 1.36%, 0.28% and 1.41%. Piecewise-β has the lowest case error; constant-β keeps growing through the test window (Figure 12, left).

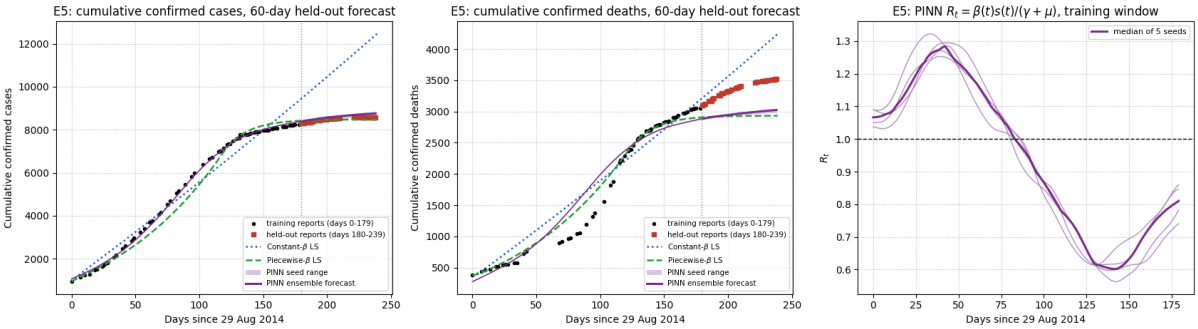


Figure 12: Experiment E5 on the WHO data: fitted to the reports in days 0–179 (29 Aug 2014 – 24 Feb 2015) and tested on days 180–239 (25 Feb – 25 Apr 2015). Left and middle: cumulative confirmed cases and deaths; right: the PINN's R_t over the training window.

2. **Deaths:** the PINN's relative error is 11.772%, against 10.694% for constant-β and 12.884% for piecewise-β. The PINN and piecewise-β forecasts lie below the reported deaths in the test window, and constant-β ends above them (Figure 12, middle). Between the last training report (23 February 2015) and 24 April 2015, WHO added 361 confirmed cases and 463 confirmed deaths. The PINN's case-fatality ratio is 0.347–0.357 across the 5 seeds.

3. **What the PINN adds:** besides the forecast, it gives R_t over the training window (Figure 12, right): it starts near 1.07, peaks near 1.28 around day 40 (early October 2014), falls below 1 around day 80 (mid-November 2014), reaches about 0.6 around day 140 (mid-January 2015) and is about 0.8 at the end of training (24 February 2015).

9.5 The whole epidemic: rolling forecasts (Experiment E7)

A single held-out window gives one comparison. To test the PINN across the whole epidemic, we use rolling forecasts on the full WHO series. At each of 8 forecast origins, every 28 days from 16 January to 31 July 2015, each method is fitted to the previous 8 weeks of reports only and forecasts the next 4 weeks. The PINN (median of 3 seeds) is compared with the same two baselines, with two β segments for piecewise-β.

Table 8: Experiment E7: mean error over 8 rolling 4-week forecasts (16 Jan – 31 Jul 2015), WHO laboratory-confirmed data.

Method	Cases MAE	Cases RMSE	4-week new-case error	Deaths MAE
PINN ensemble (3 seeds)	96.8	102.7	148.2	69.7
Piecewise-β LS	106.5	114.0	167.8	42.0
Constant-β LS	120.9	129.5	192.5	50.7

On cases the PINN's mean absolute error is 20% lower than constant-β's (96.8 against 120.9), and lower at 6 of the 8 origins; against piecewise-β (106.5) it is lower at 6 of the 8 origins (Table 8, Figure 13). On deaths the PINN is less accurate than both baselines (69.7 against 42.0 for piecewise-β and 50.7 for constant-β). Death reports in this period are noisier than case reports, and the PINN's death forecast carries any error in its learned μ. The PINN's R_t is below 1 in almost every training window from January 2015, consistent with the decline of the epidemic.

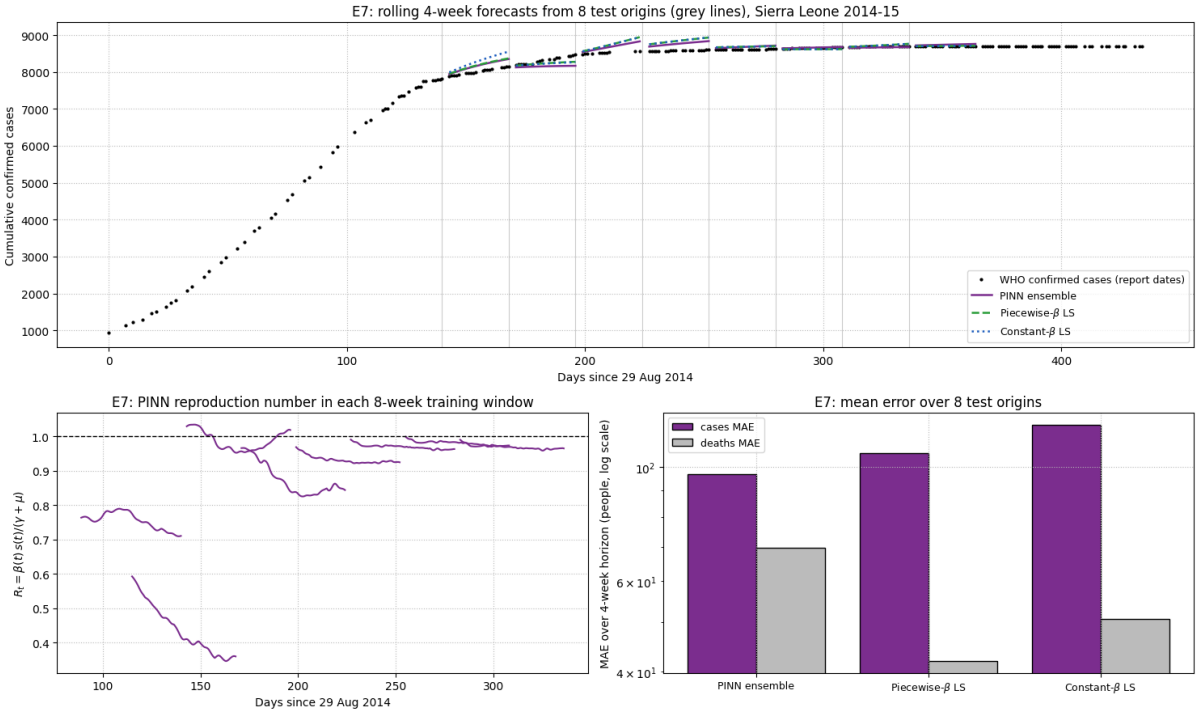


Figure 13: Experiment E7: 4-week forecasts from 8 origins on the WHO 2014–15 Sierra Leone data (top), the PINN's R_t in each training window (bottom left), and the mean errors (bottom right).

10 Ablation Studies (Experiment E6)

On the E3 problem with 5% noise (seeds 42 and 43) we changed one design choice of the PINN at a time (Table 9).

Table 9: Experiment E6: ablation on the E3 problem (5% noise, mean of 2 seeds).

Setting	$\beta(t)$ RMSE	hidden $e(t)$ rel. L_2	μ rel. error
Full method	**0.042**	**0.070**	0.054
Physics weight $w_{\text{ode}} = 10$ (instead of 1)	0.123	0.207	0.143
Adam only (8000 iterations, no L-BFGS)	0.243	0.957	0.176
No Fourier features	0.058	0.083	0.045

- **The L-BFGS stage is essential**: with Adam alone $\beta(t)$ stays almost flat (RMSE 0.243) and the hidden $e(t)$ is wrong by 96%.
- **Loss balance matters**: with $w_{\text{ode}} = 10$ the network obeys the ODE at the expense of the data, and the errors in $\beta(t)$, $e(t)$ and μ roughly triple.
- **Fourier features** lower the β RMSE by about a quarter (0.058 to 0.042).
- **Relative residuals** are decisive in the forward problem: repeating E2 with absolute residuals gives relative L_2 errors of 0.752–0.757 for s and 0.785–1.288 for the other compartments (3 seeds) instead of below 2×10^{-3}.

11 Limitations

Key limitations include:

1. **Surveillance Under-reporting and Funeral Transmission:** Ebola transmission involves traditional burial rituals and healthcare worker infections not explicitly disaggregated in a 5-compartment model.
2. **Structural Identifiability:** Inferring σ, γ, and $\beta(t)$ simultaneously from aggregate clinical counts is structurally unidentifiable; fixing σ and γ from clinical literature is essential.
3. **Forecasting assumes constant transmission:** the PINN knows $\beta(t)$ only inside its training window. Forecasts hold it at its recent value, so they cannot anticipate a new intervention or a resurgence.
4. **Sharp changes are smoothed:** the smoothness penalty and the continuous network spread an abrupt change in β over several days (E3).
5. **Deaths:** in the rolling forecasts the PINN forecasts deaths less accurately than both baselines (E7), and in E5 all three methods have death errors of 10.7–12.9%; its death forecast depends on the learned μ, and death reports were revised several times.
6. **Sensitivity to initialisation:** single PINN seeds can give noticeably different forecasts (E5: 0.28% to 1.75% case error), so an ensemble over seeds is needed.
7. **Loss balancing and cost:** results depend on the loss weights (E6) and each fit takes about 1–2 minutes on Colab's CPU, against seconds for least squares.

12 Academic Honesty and AI Acknowledgement

In compliance with Course Instruction 5:

- **AI Assistants (Claude / Antigravity / Gemini):** Utilized for checking mathematical matrix derivations, generating initial LaTeX scaffolding, and structuring PyTorch autograd loss functions.
- **Module 2 (Mathematical Analysis):** I worked through the derivations of R_0, the equilibria and their stability, and the final-size relation, and used Claude (Anthropic) to check them. It pointed out that the disease-free states are stable but not asymptotically stable, and that the old final-size check stopped too early. Claude also helped write the analysis code (growth-rate check, final-size check, LHS–PRCC) and helped me word the section clearly. I read and ran all the code; every number in the section comes from the notebook. Details are in `report/ai_use_member2.md`.
- **Module 4 (PINN) and Module 5 (E5–E7):** The PINN, its synthetic experiments (E2–E4, Module 4) and the real-data and ablation experiments (the PINN rows of E5, E6 and E7, presented by Module 5) were built on 26 October 2026 with Claude (Anthropic, via Claude Code), which designed and wrote the code (`seird_pinn.py`, `pinn_experiments.py`, `pinn_figures.py`, `data_pipeline_who.py`), ran the experiments and drafted the corresponding sections. The numbers and figures in these sections are the outputs of the final notebook run in Google Colab. Details are in `report/ai_use_member4.md`.

- **Module 1 (Biological Motivation, Formulation, Published Models and Phase-Space Analysis):** Module 1 was produced with Google Antigravity (Google's agent-based coding environment, running Gemini models), accessed through Jio. Antigravity drafted the biological motivation, the SEIRD formulation and its assumptions, the comparison of the published models (Legrand et al., Rivers et al., Althaus), the phase-plane derivation of $i(s)$, the peak formula $i_{\max}$ and the thresholds p_c and δ_c, and wrote the Module 1 code (parameters, compartment diagram, published-model simulations and table, phase-space check, threshold contour map, and the demonstration of the constant-β flaw). Details are in `report/ai_use_member1.md`.
- **Module 5 (E5–E7 and limitations):** Member 5 presents results produced with the code written by Claude (entry above). Claude also drafted the Module 5 text and, on 27 October 2026, filled in the E5, E6 and E7 numbers from the outputs of the Colab run without changing them. Details are in `report/ai_use_member5.md`.
- **Team Execution and Verification:** All equations, literature syntheses, numerical solvers, simulations, convergence studies, figure generations, and scientific interpretations were independently verified, executed, and understood across all project modules.

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